SELECTIVE ACTIVATION INTERVENTION FOR LOCALITY-AWARE MULTI-HOP FACTUAL ADAPTATION

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ABSTRACT

Factual adaptation has two objectives: make edited facts usable in multi-hop reasoning and avoid changing unrelated generations. Prompting and parameter editing can improve one objective while obscuring the other. We study VALUE-CQAG, a selective activation intervention that forms a contrast between an edited-fact prompt and the original question prompt, then gates that contrast with a content router. On a frozen 100-case, 300-question MQuAKE-CF confirmation pool, question-level generation rises from 2.00% to 38.67% on Qwen2.5-7B and from 4.00% to 51.33% on Llama-3.1-8B. Explicit facts remain a stronger accuracy control (95.0% and 93.0%), so our claim is selectivity rather than state-of-the-art accuracy. On a separate Qwen 10,000-pair audit, the routed system achieves 98.49% exact and 99.62% semantic locality with a 1.52% false-positive route rate. Failure analysis separates routing misses from old-answer and multi-hop entity errors. The result is a reversible adaptation path with a remaining bottleneck: multi-hop composition after a successful route.

Index Terms— large language models, factual adaptation, activation intervention, locality, multi-hop reasoning

1. INTRODUCTION

Factual adaptation is not a single objective. A useful system must both propagate an edited fact through a multi-hop answer and abstain on questions that do not depend on that edit. These requirements are easy to conflate: a prompt containing the new fact can improve answerability, while a parameter or activation change can also alter unrelated generations. The distinction matters for rapid, reversible adaptation, where an edit should be removable without producing a globally changed model.

Existing approaches solve different parts of this problem. Parameter-editing methods locate and rewrite factual associations in feed-forward layers [1–4]; editable-network and learned-editor approaches instead amortize the update operation [5–7], while recent surveys organize the broader design space and its evaluation pitfalls [8]. External memories store an edit for later retrieval [9]; retrieval or in-context methods supply explicit evidence at inference time [10, 11]. MQuAKE shows why single-hop success is insufficient: the edited fact must survive composition with other entities [12]. A single accuracy number therefore hides whether the system missed the edit, retained the old answer, or selected the wrong entity after receiving the edit.

We study VALUE-CQAG as a controlled answer to that question. It forms an answer-independent contrast between an edited-fact prompt and the original question prompt in the model's value-projection channel. A fixed content router decides whether to apply that contrast; rejected queries follow the original prompt path, while accepted queries receive a bounded prompt-only intervention before the same frozen decoder. This decomposition makes scope and effect measurable separately. Figure 1 summarizes the signal path.

Our contributions are:

- A parameter-free, gated activation operator that separates edit direction from edit scope and can be bypassed query by query.
- A controlled cross-model evaluation on Qwen2.5-7B and Llama-3.1-8B against explicit-fact, retrieval-plus-ICL, and MEMIT controls on the same frozen confirmation protocol.
- A 10,000-pair end-to-end locality audit and a failure decomposition showing that routing errors are distinct from downstream multi-hop entity errors.

2. METHOD

2.1. Problem setup

Let $\mathcal{E} = \{(s_i, r_i, o_i^{\mathrm{new}})\}_{i=1}^{K}$ be a set of edited subject–relation–object facts and let q be a question whose answer may require several facts. The base model f_θ is frozen. We seek a conditional intervention $\Delta(q, \mathcal{E})$ such that

$$\hat{y} = f_\theta(q; \Delta(q, \mathcal{E})) \tag{1}$$

is consistent with the edited facts, while $f_\theta(q; 0)$ is retained for unrelated questions. Parameter editing instead produces one model $f_{\theta+\Delta\theta}$ for all subsequent inputs. Our target is therefore a conditional computation: change the signal only when the query is related to the edit.

2.2. Activation target and gated operator

For each stored source question $s \in \mathcal{S}_{\mathcal{E}}$, let $v_l^0(s) \in \mathbb{R}^{d_l}$ be the final-token output of the self-attention value projection at layer l under the original prompt, and let $v_l^+(s, \mathcal{E})$ be the final-token output when the same question is preceded by the edited-fact instruction. The two prompts may have different lengths; using their respective final prompt tokens gives an explicit semantic anchor and avoids token-by-token alignment. We form the answer-independent direction

$$d_l(s, \mathcal{E}) = v_l^+(s, \mathcal{E}) - v_l^0(s). \tag{2}$$

No target answer or generated completion enters d_l. For a target query q, the router selects a source template $s^*(q)$ and returns

$g(q, \mathcal{E}) \in \{0, 1\}$. Let $V_l^0(q) \in \mathbb{R}^{T_q \times d_l}$ be the target prompt's value-projection output and $b_t = \mathbb{I}[t > T_q - w]$. At selected layers $\mathcal{L}$, the same direction is broadcast over the final w target-prompt positions:

$$\widetilde{V}_l(q)_{t,:} = V_l^0(q)_{t,:} + g(q, \mathcal{E})\, \alpha b_t d_l(s^*, \mathcal{E}). \tag{3}$$

For Qwen, the confirmation configuration uses $\mathcal{L} = \{20, 22, 24, 26\}$, $w = 16$, and $\alpha = 5$; the Llama confirmation uses the model-specific set $\mathcal{L} = \{16, 20, 24, 28\}$ with the same w and α. Both settings were selected on development cases and frozen before confirmation. Equation 3 is applied only during prompt encoding; weights, caches after the prompt pass, and later autoregressive states are not edited. Direct explicit-fact prompting remains an accuracy upper control. The purpose of the contrast is different: it exposes an edit through a removable, query-gated activation path while decoding from the original question prompt.

The construction is related to recent activation steering and representation intervention methods [13–16], which show that useful task information can be exposed or shifted in hidden states without updating model weights. It differs in two ways: the direction is obtained from an explicit-fact contrast rather than a learned human-preference direction, and a content router gates the direction on a query-by-query basis.

2.3. Content-token routing

Each edit stores the source-question templates used to construct its directions. For a query q, the router lowercases text, extracts alphanumeric tokens, removes a fixed English stop-word list, and selects the source question with maximum Jaccard overlap:

$$s^*(q) = \arg\max_{s \in \mathcal{S}_\mathcal{E}} J(q, s), \quad \rho(q, \mathcal{E}) = J(q, s^*), \tag{4}$$

where $J(q, s) = |T(q) \cap T(s)|/|T(q) \cup T(s)|$. The gate is

$$g(q, \mathcal{E}) = \mathbb{I}[\rho(q, \mathcal{E}) \geq \tau], \qquad \tau = 0.35. \tag{5}$$

The threshold is fixed on development data and is not tuned on the confirmation pool. A router miss is counted separately from a generation error after a successful route. This abstention mechanism motivates reporting coverage and generation separately rather than hiding both behind one accuracy score.

2.4. First-order effect and locality identity

For a fixed answer position, let $m(q) = z_{y_{\text{new}}}(q) - z_{y_{\text{old}}}(q)$ be the new-versus-old logit margin, and let $\mathcal{W}_q$ index the final w prompt positions. A local expansion around the unmodified activations gives

$$\Delta m \approx \alpha g \sum_{l \in \mathcal{L}} \sum_{t \in \mathcal{W}_q} \left\langle \nabla_{V_l[t,:]} m, d_l(s^*, \mathcal{E}) \right\rangle + O\left(\alpha^2 w \sum_l \|d_l\|_2^2 \right). \tag{6}$$

The equation is diagnostic, not a global guarantee: a useful contrast must align with the local answer-margin gradient, whereas an unrelated equal-norm direction has no systematic alignment under an isotropic null model. Multi-token generation can still fail after a favorable local change, which motivates the separate old-answer and entity-error audit.

The router also gives an exact locality decomposition. For an unrelated query u, let $\phi = \Pr[g(u, \mathcal{E}) = 1]$ be the false-positive route rate and let $\eta = \Pr[\text{output changes} \mid g = 1]$. Because rejected queries bypass Equation 3,

$$L_{\text{exact}} = 1 - \phi\eta. \tag{7}$$

On the 10,000-pair audit, $\phi = 152/10{,}000 = 1.52\%$ and $\eta = 0.9934$, yielding $L_{\text{exact}} = 98.49\%$. Equation 7 explains why a high router rejection rate is necessary but not sufficient: a false route is highly likely to alter the output.

2.5. Two-stage prompt-pass implementation

Figure 1 shows the two stages. Stage 1 constructs $d_l(s, \mathcal{E})$ for each source template with frozen passes; independently, the text router selects $s^*(q)$. Stage 2 either forwards the target prompt unchanged or broadcasts the selected direction over its final w positions before the shared frozen decoder. Extending the intervention through the first two generated tokens reduces question accuracy by 4.00 points on Qwen and 7.00 points on Llama, supporting the prompt-only design.

3. RELATED WORK

Knowledge editing has been studied through localized parameter updates such as ROME and MEMIT [2, 3], learned editors [6, 7], and external edit memories such as SERAC [9]. Localization and specificity studies show that successful edits can remain difficult to predict and can introduce unintended behavior [17, 18]. MQuAKE specifically exposes failures that appear when several edited facts must be composed [12]. Our intervention instead controls the activation signal presented to a frozen decoder.

Activation steering methods add directions at inference time, including unsupervised activation addition [13], inference-time interventions for truthful behavior [14], and function-vector analyses of in-context computation [15]. Work on latent knowledge and relation decoding suggests that internal representations can carry answer-relevant relational information even when final generation is unreliable [16,19]. These methods motivate treating hidden states as a controllable signal, but generally apply a direction whenever a prompt is evaluated. Our contribution combines an explicit-fact contrast, a lexical gate, and an end-to-end locality audit. Retrieval-augmented generation remains a strong accuracy control [10].

4. EXPERIMENTAL PROTOCOL

4.1. Models and data

We evaluate Qwen2.5-7B-Instruct and Llama-3.1-8B-Instruct. All principal comparisons use the same frozen 100-case MQuAKE-CF confirmation pool, containing 300 questions (three per case); MQuAKE is designed to test composition rather than isolated single facts [12]. The pool is separate from a reserved set that is not used in this study. For parameter editing, five deterministic unrelated questions per edited case form a 500-output locality check; the routed audit is reported separately below.

4.2. Baselines and metrics

We compare four adaptation modes: (i) the unmodified base model, (ii) explicit facts placed in the prompt, (iii) a transparent retrieval-plus-ICL control inspired by IKE [11], and (iv) MEMIT [3] implemented with EasyEdit [20]. The retrieval corpus contains 500 non-overlapping development cases and uses MiniLM semantic retrieval with eight demonstrations. This is an IKE-style control, not a claim of an unmodified official IKE reproduction.

We report question-level generation accuracy, case-level generation accuracy (all questions in a case correct), new-answer preference, and exact-output locality. A generation hit requires normalized

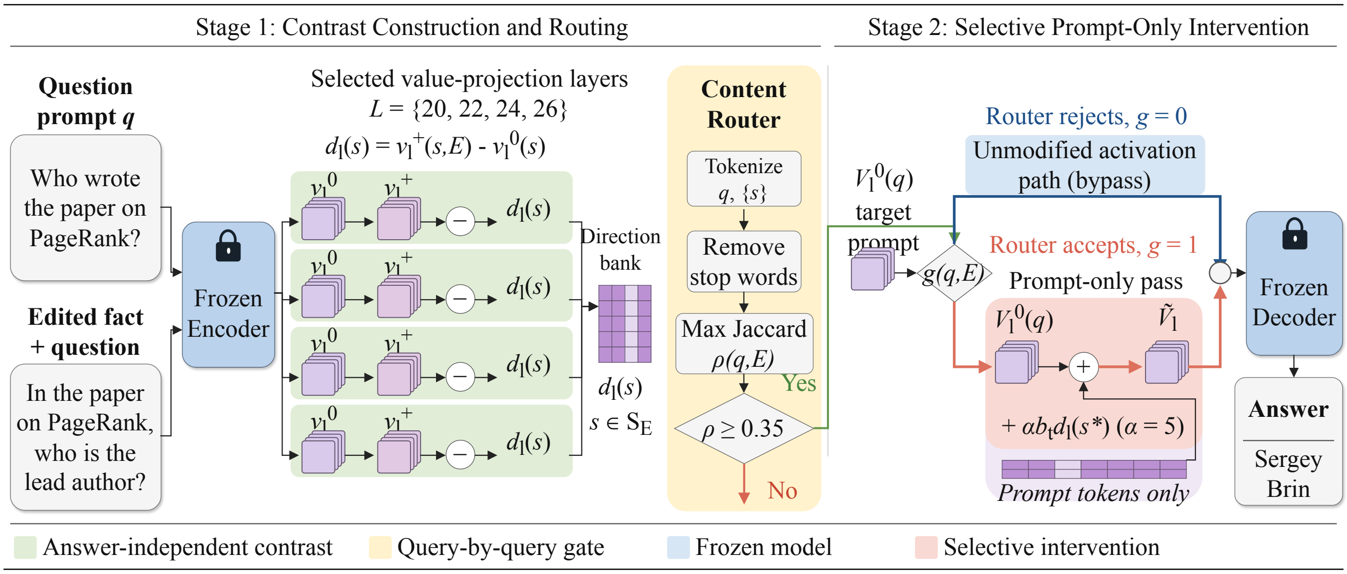

Fig. 1. Two-stage VALUE-CQAG pipeline (Qwen layers shown; Llama uses $\{16, 20, 24, 28\}$). Stage 1 subtracts final-token value vectors for each stored source template s to obtain $d_l(s, \mathcal{E})$; the text router independently selects $s^*(q)$. In Stage 2, rejected queries bypass the edit, while accepted queries receive the selected direction over their last w prompt positions before frozen decoding.

exact match to the new answer or a recorded alias. The routed locality audit crosses 100 edited cases with 100 unrelated questions, yielding 10,000 pairs. Exact locality requires an unchanged output string; semantic locality requires preservation of the task's answer-equivalence status. The MEMIT locality protocol uses 500 fixed unrelated outputs and is not pooled with the routed audit. Confidence intervals are paired case-bootstrap intervals where reported.

4.3. Reproducibility

For VALUE-CQAG, models use 4-bit NF4 loading with fp16 compute, greedy decoding, and at most 24 new tokens; a generation hit is normalized exact match to the new answer or a recorded alias. For MEMIT, second-moment statistics use the local ‘wikimedia/wikipedia‘ 20231101.en mirror, 100,000 samples, float32, and layers 4–8 of ‘mlp.down_proj‘. Due to the 24-GB RTX 3090 memory limit, MEMIT models are loaded in fp16; the covariance linear system is solved in CPU double precision and only the resulting update is moved to the GPU. The scripts, hparams, frozen IDs, and raw JSON artifacts accompany this paper draft.

5. RESULTS

5.1. Cross-model comparison

Table 1 states the paper's central trade-off. VALUE-CQAG substantially improves over the base model on both architectures, but explicit facts and retrieval remain stronger on answer accuracy. The contribution is therefore controlled adaptation: the method changes the original prompt path only when the router accepts the query, rather than claiming to replace a long evidence prompt on raw accuracy.

Relative to the base model, the paired question-level gain is +36.67 percentage points for Qwen (95% case-bootstrap CI [29.33, 44.33]) and +47.33 points for Llama (95% CI [39.33, 55.33]).

Table 1. Question-level and case-level generation accuracy on the same frozen 100-case (300-question) confirmation pool. Retrieval+ICL† was run on Qwen only, using eight demonstrations from 500 development cases.

Model	Method	Question	Case
Qwen2.5-7B	Base	2.00%	0.0%
	VALUE-CQAG	38.67%	18.0%
	Explicit facts	95.00%	90.0%
	Retrieval+ICL†	95.00%	93.0%
Llama-3.1-8B	Base	4.00%	0.0%
	VALUE-CQAG	51.33%	32.0%
	Explicit facts	93.00%	87.0%

Retrieval-plus-ICL reaches 95.0% question-level and 93.0% case-level accuracy on Qwen, but its exact and semantic locality are only 25.0% and 61.4%. Thus, VALUE-CQAG is a selective, lower-interference alternative rather than the strongest accuracy baseline.

Both models routed 281/300 questions (93.67%; 19 misses). Prompt-only 95% CIs were [31.33, 46.33] for Qwen and [43.67, 59.33] for Llama, indicating that the larger Llama gain was not caused by different router coverage.

5.2. Locality and parameter editing

The end-to-end routed audit on Qwen gives a 1.52% false-positive route rate, 98.49% exact-output locality, and 99.62% semantic locality. Conditional on a false route, exact locality falls to 0.66%, showing that the router is the main safety boundary rather than evidence that the injected direction is intrinsically local. The audit records 37 newly corrupted originally-correct outputs and only one newly fixed originally-incorrect output.

Table 2 compares the two MEMIT runs. The method has lower generation accuracy than VALUE-CQAG and substantially lower lo-

Table 2. EasyEdit MEMIT on the frozen 100-case pool. Locality uses 500 unrelated outputs.

Model	Pref.	Q-gen.	Case	Loc.	Time (s)
Qwen2.5-7B	16.0%	16.0%	8.0%	85.2%	52.41
Llama-3.1-8B	18.0%	14.33%	5.0%	73.4%	29.64

Table 3. Completed auxiliary evaluations (V-CQAG denotes Value-CQAG). Sample counts are in parentheses. The first two rows report three-seed mean±standard deviation; the others are single runs. "Loc." is mismatched-direction locality for V-CQAG and exact-output locality for ROME.

Setting (n)	Q-gen.	Case	Loc.
Qwen CF, V-CQAG (30×3)	27.78±2.22	12.22±1.92	80.00±14.53
Qwen T, V-CQAG (16×3)	21.53±2.41	0.00	64.58±7.22
Qwen2.5-3B CF (30)	3.33	0.00	–
Llama CF, q-specific (80)	42.92	28.75	–
Qwen CF, ROME (80)	19.17	13.75	80.75

cality on Llama. These results indicate that a weight edit can provide a persistent change, but persistence is not equivalent to selective behavior on unrelated questions.

5.3. Failure decomposition

Among 300 Qwen questions, 116 are correct, 133 are other-entity or reasoning errors, 31 retain the old answer, 19 are router misses, and one is a refusal. Llama yields 154 correct, 96 other-entity or reasoning errors, 30 old-answer residuals, 19 router misses, and one refusal. The identical router-miss count across models separates coverage from the downstream generation bottleneck: after a successful route, multi-hop entity selection remains the dominant failure mode.

5.4. Robustness and auxiliary evaluations

Table 3 reports auxiliary evaluations. Their case pools and filtering rules differ from the confirmation protocol, so they are used only to test whether the observed behavior is consistent across settings, not as pooled evidence.

The explicit-fact control reaches 95.0% / 90.0% question/case accuracy on Qwen and 93.0% / 87.0% on Llama. On the earlier 30-case split it reaches 78.89% on Qwen2.5-7B and 74.44% on Qwen2.5-3B, confirming that the evaluation is solvable and that the remaining gap is methodological rather than an impossible target.

Coverage versus conditional gain. Both models accept 281/300 questions (93.67%), so their different gains reflect decoder use of the same signal rather than different gate coverage. The 19 misses are a measurable recall ceiling for lexical routing.

Residual errors. After routing, entity/composition errors dominate (133 Qwen; 96 Llama), while old-answer residuals account for 31 and 30 cases. Lowering τ may recover paraphrases but increases false routes and directly reduces locality through Equation 7; improving composition is therefore safer.

Baseline interpretation. Explicit facts and retrieval show the task is solvable, while retrieval's 25.0% exact locality and Llama MEMIT's 73.4% locality expose the interference cost. Value-CQAG trades peak accuracy for a reversible, query-level scope decision.

5.5. Ablations and efficiency

The development and probe runs provide directional evidence for the operator in Equation 3. A single token gave 25% new-answer preference on a 12-question probe, whereas a 4–16-token window reached 75%. On the 30-case CF split, one layer (22) gave 46.67% preference and 76.67% mismatched locality; three layers (22,24,26) gave 66.67% preference but 63.33% locality. Continuing the injection for two decoding steps reduced confirmation question accuracy from 38.67% to 34.67%; a full token trajectory reached only 17.0% on a development validation. These are ablations, not additional independent confirmation claims.

In an 80-case audit, direction construction, base generation, and steered generation averaged 2.273 s/edit, 0.388 s/question, and 0.924 s/question. The 100-case locality audit required 3.584 s/edit to construct three direction sets; rejected pairs reused cached base outputs. Retrieval-plus-ICL required 4.655 s/case for retrieval and generation over three questions. These different workloads permit descriptive timing, not a matched speed claim. No-router exact locality was 3.33%. Two-step continuation reduced question accuracy by 4.00 points on Qwen (95% CI [–8.67, 0.33]) and 7.00 points on Llama ([–11.00, –3.33]).

6. DISCUSSION AND LIMITATIONS

The evidence supports a focused conclusion. Conditional activation intervention improves multi-hop factual adaptation on two instruction-tuned models without changing their parameters, while the router yields high system-level locality. The same audit also shows the boundary of that claim: after a false route, exact locality falls to 0.66%, and after a successful route, multi-hop entity selection remains the dominant failure mode. Retrieval is a strong accuracy control; VALUE-CQAG is useful when selective, reversible behavior is more important than maximizing answer accuracy.

The confirmation pool contains 100 cases and is not a broad pretraining-scale benchmark. The router uses lexical overlap with stored source-question templates and can miss paraphrases. The five-question parameter-editing locality protocol and 10,000-pair routed audit measure related but different quantities. Finally, the MEMIT comparison uses a documented hardware compatibility path (fp16 loading and CPU double solve) on a 24-GB GPU; it is not presented as an unmodified official MEMIT run. Future work should test relation-macro accuracy, independent seeds, semantic routing, matched end-to-end resource budgets, and a learned locality-regularized router. These limitations bound the claim; they do not change the observed separation between routing coverage and downstream composition.

7. CONCLUSION

We presented VALUE-CQAG, a selective activation intervention for multi-hop factual adaptation. Its central design choice is to separate what changes from where it changes: an explicit-fact contrast supplies the direction, while a content router controls exposure and rejected queries retain the original path. Across Qwen and Llama, this improves over the base model but does not surpass explicit prompting on raw accuracy; the value of the method is the auditable locality boundary and the resulting failure decomposition. The next technical target is clear: replace lexical routing and improve multi-hop composition without sacrificing that boundary.

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Compliance with Ethical Standards

This study uses publicly available benchmark cases and pretrained models; it involves no human subjects, user data, or private or personally identifiable information. The evaluation is automated and reports aggregate outcomes. The reserved 60-case pool remains sealed and was not read or evaluated during this study.

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